Resonant Tube

1 Aim

To measure the speed of sound in air at room temperature, using the resonance of an air column.

2 Introduction

The resonant column apparatus consists of a long metal pipe/column mounted on a vertical board (with a marked metre scale alongside) and a rubber tubing connecting the column to a beaker containing water. The water level in the column can be adjusted by lowering/raising the beaker using the attached clamp. The resonance tube can be leveled using the leveling screws provided at the bottom. Tuning forks of different frequencies are provided to achieve different resonant frequencies. When a tuning fork of a particular frequency is made to vibrate over the open end of the partially filled (with water) resonant column, sound waves propagate in the column which are reflected back from the surface of the water and form standing waves due to interference.

3 Apparatus in the lab and procedure

1. Strike a tuning fork on the rubber pad provided, hold it over the open end of the column and adjust the water level in the column till you obtain a resonance. Note down the corresponding meter scale reading of the length of the column above the water level. Once you have found the approximate position of the resonance point, make several trials by running the water surface up and down and make an estimate of the error involved in locating the point of resonance. This is the first resonance position (say L_1). Confirm this resonance position by taking four readings: two for when level of water is falling and two for when water level is rising.

Figure 1: Resonant tube apparatus.

2. Lower the water level position so that is around three times the length L_1 . Repeat the above procedure to get a second resonance position (say L_2). Again obtain the mean of four readings (two for water rising and two for water falling). Repeat the experiment and obtain the two mean resonance positions for a different tuning fork frequency.
3. Use a set square to measure the lower meniscus of the water level.
4. Use a thermometer to measure the temperature of air.
5. Calculate the velocity of sound in air at room temperature v. by plugging in the values of the tuning fork frequency ν and the resonance position lengths L_1 , L_2 into the formula: $v = 2\nu(L_2 - L_1)$

3.1 Points to Ponder

1. How does a flute work?